11.2 Areas of Circles and Sectors

Name _____ Period ____ Date _____

Geometry 11.2 (30-minute plan) · Student sheet

I can…

- find the area of a circle and use it to find a radius.
- find the area of a sector of a circle.
- find the area of a shaded region made of circles and polygons.

Vocabulary · fill in the missing word

Sector of a circle — the region bounded by two radii of the circle and their _____ arc

Area of a circle — the number of _____ units inside the circle: $A = \pi r^2$

Part 1 · Notes and examples (fill in each note, then work the example)

Area of a circle: A = _____

Example 1

Find the area of each circle. Round to the nearest hundredth.

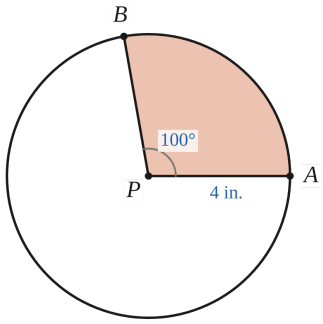
a. radius 2.5 centimeters

b. diameter 10 inches

Area of a sector: area of sector APB = (m(arc AB) ÷ 360°) · _____

Example 2

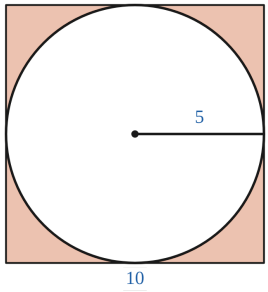
Find the area of the shaded sector and of the unshaded sector. Round to the nearest hundredth.



Shaded regions: area of the shaded region = area of the whole figure _____ the area of the part that is not shaded.

Example 3

A circle is inscribed in a square with sides of 10. Find the area of the shaded region. Round to the nearest hundredth.



Part 2 · Partner practice

Partner practice

A circle with a diameter of 8 is inscribed in a square with sides of 8. Find the area of the region inside the square but outside the circle. Round to the nearest hundredth.